The design of `GaugeTheory/`

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This version. It covers all five layers and the route to the target theorem. It was written after a review of the gauge-theory folder of PhyslibAlpha (77 files, 27 545 lines), of the Standard Model tree built on it (119 files, 54 772 lines) and of the branch of PR #2011, at Physlib `master` commit `b512ca5` (10 November 2026). The order of work agreed on 10 November stands: layer by layer, layers 1 and 2 first.

Statements about the code are marked. EXISTS: in the tree today. CHECKED: compiled in Lean against the branch of PR #2011, in one of the three prototype files in `prototypes/` of this repository (1 215 lines, no `sorry`; Appendix A) or, for the import fixes of §3, as a copy of the file with its import changed. PROPOSED: a design choice for the thread to decide. TO DO: follows from the design and is not written yet.

What the review found.

1. **The layers hold.** With every file assigned to a layer by its content, imports go downward only, except in two files, once the coefficient extraction of PR #2011 and two import fixes are in (§3).
2. **The two compatibility conditions can be laws.** They hold for every matter field the library can build: with one evaluation law on the factors of a gauge group every row of every table satisfies both, and restriction and direct sums preserve them (CHECKED, §5.2).
3. **The covariant reduction theorem is general.** For any field content over free gauge data, an expression invariant under all jets of gauge transformations is a polynomial in covariant derivatives (CHECKED, §8). Until now it was proved only for the twelve field families of the Standard Model.
4. **Equivalent descriptions have the same invariants.** A renaming of species together with isomorphisms of their matter fields induces an isomorphism of the algebras that carries invariants to invariants (CHECKED, §7.3).
5. **The Standard Model classification stands beside the general layers, not on them.** The two share only the algebra. Three comparison lemmas and one equivalence state the theorem on the card (§9).

1 Purpose

`GaugeTheory/` is the symbolic framework for the local content of a gauge theory: the algebra in which a Lagrangian is written, the gauge and Lorentz actions on it, and the invariants of a given mass dimension. Fields enter only through their jets at one point, the formal symbols $\partial_s \psi_\alpha$.

The target theorem. For the Standard Model, given as a card, the gauge- and Lorentz-invariant elements of its local field algebra of mass weight at most 8 (mass dimension at most 4) are spanned by an explicit list: the constant, $H^\dagger H$ and the terms of dimension four. It classifies invariant polynomials over $\mathbb{C}$, with no reality condition and no quotient by total derivatives or equations of motion. A *card* is the file a model writes: its gauge group as a list of factors, and a table of its fields with their charges and numbers of generations (`PhyslibAlpha/Particles/StandardModel/Basic.lean`).

What the library must do.

1. **Describe a theory from its card.** The list of factors and the table produce the gauge data, the matter fields, the connection and the field data. *Layers 1 to 3;* EXISTS.
2. **Write and classify its local expressions.** The algebra of the symbols carries the gauge and Lorentz actions, the total derivative and the mass-weight grading; the invariants of a given mass dimension come as a spanning list. *Layers 4 and 5; the general part* EXISTS *or* CHECKED*, the classification proved for the Standard Model on hand-built data* (*§9*).
3. **Relate theories to one another.** Equivalence, embedding, projection, and restriction along a change of gauge group. *§10.*
4. **Reality, and what comes after.** Real fields and a star; invariants modulo total derivatives and equations of motion; symmetry breaking; flavour; higher mass dimension. *After the deadline* (*§12*).

Deadline. Early December 2026, with the end of the semester as the hard deadline.

2 Design rules

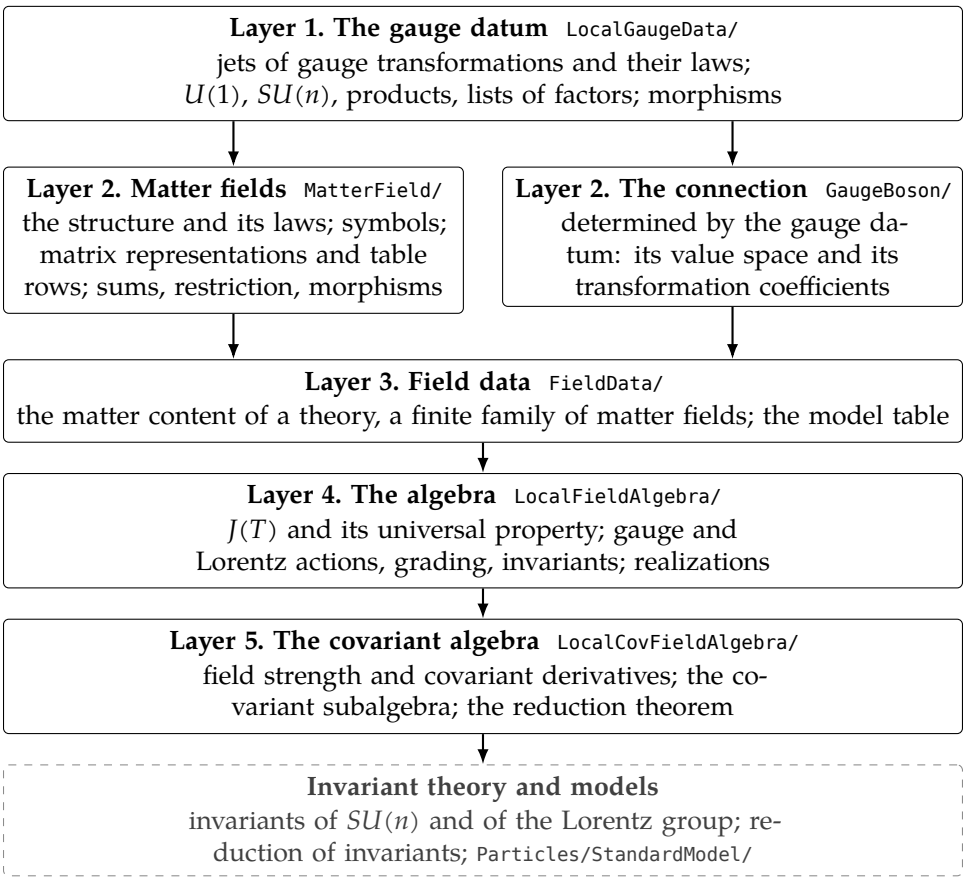
1. **Data in, theorems out.** Every layer takes a datum and returns constructions and theorems. A concrete theory contributes only its card.
2. **Laws belong to the datum.** A property that every instance has is a field of the structure, stated once; a theorem should not carry it as a hypothesis. A property that only some instances have is a class, like `Faithful` and `Free`. A law of one layer is usually the law of the layer below read on new objects: the gauge datum says that at the base point a jet acts on the Lie algebra through its value, and a factor, a matrix representation and a matter field should say the same of their own actions (§5.2).
3. **Symbols are covectors.** A field is described by its value space. The generators of the algebras are linear functions on jets of values, so they pull back along maps of value spaces: a map of theories $T \to T'$ gives a map of algebras $J(T') \to J(T)$.
4. **Statements live in an arbitrary algebra.** A law of symbols, or a classification, is stated for families of symbols in any algebra B that carries the actions. $J(T)$ is the universal such algebra, and an equivariant algebra map out of it is a *realization* of T in B. A result proved once then applies to another description of the theory, to a sub-theory and to a quotient.
5. **Morphisms, not equality.** A matter field contains its value space as a type, so two isomorphic fields are never equal in Lean. Theories are compared by morphisms.

3 The layers

The library has five layers. Layers 1 to 3 are data: what a theory is made of. Layers 4 and 5 are the algebras built from the data. The names `FieldData`, `LocalFieldAlgebra/` and `LocalCovFieldAlgebra/` were agreed on 3 November; in the tree they are still `GaugeFieldData`, `LocalField/` and `LocalCovField/`.

Two kinds of field. Matter fields and the connection are separate notions (AGREED, 2–3 November). A matter field is data a theory chooses. The connection is not chosen: its value space CoVector $\otimes$ $\mathfrak{g}$ and its affine transformation law follow from the gauge datum. A closed type `Species | matter M | gauge` combining the two was tried in PR #2011 and dropped on 3 November: every general lemma split into the two cases, and the matter case only restated existing results.

The connection has a part in every later layer. Its own algebra of symbols depends only on the gauge datum, is built before $J(T)$, and is one tensor factor of $J(T)$. Its field strength and the reduction theorem for the gauge field alone are likewise built before the covariant algebra of a theory. These parts are in `GaugeBoson/` today. They belong to layers 4 and 5, not to layer 2. The covariant derivative of a matter field, in `MatterField/` today, belongs to layer 5.



The assignment. Every file is assigned to a layer by its content. Sizes are on `master`; paths are under `PhyslibAlpha/ClassicalFieldTheory/GaugeTheory/`.

layer	files	lines	where it is today
1. gauge datum	11	3 349	`LocalGaugeData/` without its $SU(n)$ invariant theory; the structure itself (431 lines) is in `Physlib/`
2. fields	14	4 303	`MatterField/`: structure, `Charge`, `TransformsIn`, `JetComponent/`, `MatrixRep/` without `Table`, `Prod`, `Pi`. `GaugeBoson/Basic`. The jet toolkit `LocalField/Jet`, `JetRep`
3. field data	8	2 475	`GaugeFieldData/`; `MatterField/MatrixRep/Table`
4. algebra	19	5 948	`LocalField/` without `CovariantDeriv`; `GaugeBoson/LocalGauge/` without `FieldStrength`; `GaugeBoson/Realization/Basic`
5. covariant algebra	17	8 334	`LocalCovField/`; `LocalField/CovariantDeriv`; `MatterField/CovariantDeriv`, `InfinitesimalAction`, `LorentzCovariantDeriv`; the rest of `GaugeBoson/Realization/`; `GaugeBoson/LocalGauge/FieldStrength`; `GaugeBoson/LocalGaugeCov/`
invariant theory	8	3 136	`LocalGaugeData/SU/Invariants/`, `GellMann`, `Adjoint`, `TensorSpecies`

The import check. The script `tools/check_layers.py` of this repository holds the assignment and lists the imports that go from a lower layer to a higher one. On `master` there are four. Two are in layer 2: the structure `MatterField/Basic` and the definition `MatterField/TransformsIn` reach the covariant derivative. PR #2011 removes them, by moving the Taylor coefficients of a representation to `MatterField/`

`RepCoeff` and the law of the infinitesimal action to `InfinitesimalAction/Basic`. It leaves two stale imports of its own, and both files compile with the import narrowed (CHECKED): `MatrixRep/Basic` needs only `InfinitesimalAction/Basic`, and `JetComponent/TransformsIn` needs nothing from `CovariantDeriv`. A third narrowing is not a layer violation but trims what the structure depends on: `InfinitesimalAction/Basic` compiles with `RepCoeff` and `LocalGaugeData/AdjointCoeff` in place of `MatterField/TransformsIn` (CHECKED). The other two upward imports are in layer 4 and remain: `LocalField/Realization` and `GaugeBoson/LocalGauge/Realization` import the covariant towers, because each also states what a realization does to them. Splitting those statements off can wait for layers 4 and 5.

Directories. No file has to move for the layers to hold; the table is the assignment and the imports are the test. Three things should move all the same. The jet toolkit does not depend on a gauge group and goes below `MatterField/`. The model table and its compilation go with the field data, the rows staying in layer 2. The $SU(n)$ invariant theory leaves `LocalGaugeData/` for the classification. The connection's algebra and the covariant derivative move when layers 4 and 5 are upstreamed.

4 Layer 1: the gauge datum

The question it answers. A Lagrangian density at a point x depends on the fields and finitely many of their derivatives at x. A gauge transformation $g(x)$ therefore matters only through its Taylor series at that point, its *jet*. This layer describes these jets and nothing else.

An example. For $U(1)$, a gauge transformation is $u(x) = e^{i\theta(x)}$. Its jet is the power series of u in the coordinates $x_0, \dots, x_3$, a unitary element of $\mathbb{C}[[x]]$. Its value at the base point is $u(0) \in U(1)$. Its *Maurer–Cartan form* $\omega_\mu = i(\partial_\mu u)u^{-1} = -\partial_\mu\theta$ is the shift of the gauge field, $A_\mu \mapsto A_\mu + \omega_\mu$. For $SU(n)$ everything is the same with $n \times n$ matrices of power series.

The datum (EXISTS, in `Physlib/`). `LocalGaugeData` G_0 $\mathfrak{g}$ `GJ` $\mathfrak{g}_J$ has four types, the gauge group G_0, its Lie algebra $\mathfrak{g}$ and their jet versions G_J and $\mathfrak{g}_J$, and the maps a transformation law uses:

- evaluation at the base point and constant jets, $G_J \leftrightarrows G_0$ and $\mathfrak{g}_J \leftrightarrows \mathfrak{g}$;
- derivatives ∂_μ and multiplication by coordinates x_μ on $\mathfrak{g}_J$;
- the adjoint actions, of G_J on $\mathfrak{g}_J$ and of G_0 on $\mathfrak{g}$;
- the Maurer–Cartan form $\omega_\mu(U)$.

The rules are those of the example. The derivatives commute, are derivations of the bracket and kill constants. Ad_U preserves the bracket, obeys a Leibniz rule, and at the base point is $\mathrm{Ad}_{U(0)}$. The Maurer–Cartan form vanishes on constant jets, is a cocycle, $\omega(UV) = \omega(U) + \mathrm{Ad}_U\,\omega(V)$, and is flat, $\partial_\mu\omega_\nu - \partial_\nu\omega_\mu + [\omega_\mu, \omega_\nu] = 0$, where $[A, B] = i(AB - BA)$ on hermitian generators. Everything later is proved from these rules, so it holds for every gauge group at once.

What is proved from the rules (EXISTS). Iterated derivatives ∂_s along a multiset of directions. The Taylor coefficients of Ad_U (`AdjointCoeff`), which is how a gauge jet moves the derivatives of a field. The symmetrised Maurer–Cartan data (`MaurerCartan`). The jets that agree with the identity to order n (`truncationKer n`), normal subgroups decreasing in n; the zeroth is the *pure jets*, those with trivial value.

Two classes (EXISTS). The rules do not say that G_J consists of honest Taylor series. `Faithful`: a Lie-algebra jet is determined by the base-point values of its iterated derivatives, and a gauge jet with vanishing Maurer–Cartan form is constant. `Free` adds that every family of Taylor data is realised, by a Lie-algebra jet and, through the Maurer–Cartan form, by a pure jet; the symmetrised Maurer–Cartan data are then free coordinates on the pure jets. `Free` is what the reduction theorem of layer 5 needs.

Concrete groups (EXISTS). A *matrix presentation* (`MatrixJets`) says that the four types sit inside matrices of jets and every map is the matrix one. From it the rules and `Faithful` are proved once, and `Free` is reduced to three conditions on the carriers. $SU(n)$ (`su n`) and $U(1)$ (`u1`) fill in a presentation and are free. Products (`prod`) and lists of factors (`ofFactors`) give any group such as $SU(3) \times SU(2) \times U(1)$, written `[.SU 3, .SU 2, .U1]`, and freeness passes to them.

Morphisms and factors (EXISTS). A *morphism* (`Hom`, in `Physlib/`, with identities and composition) is a map between two data respecting all the maps above; the projections of a product are morphisms (`Hom.fst`, `Hom.snd`). Charges are given one factor at a time. A *factor* (`U1Factor`, `SUFactor`) records how the gauge jets project onto one $U(1)$ or $SU(n)$, and pulls back along a morphism (`Factor.comap`). That is how the factors of a list are obtained from those of its pieces.

The evaluation law (PROPOSED, CHECKED). A factor has three of the four maps a morphism has: `u` on gauge jets, φ on the gauge algebra and φ_J on its jets. It has no law tying the unitary jet `u U` to the value of U. A factor can therefore be twisted by a character of the pure jets, and the representations built from it need not act through the value of a jet at the base point. `MatrixJets` has the missing law (`toMat`$_0$`_eval`); the factors drop it. Proposed: one more law on `U1Factor` and `SUFactor`, that the unitary jet of a jet with trivial value has constant term 1. It holds for the canonical factors of `u1` and `su n`, and it passes through `Factor.comap` because a morphism commutes with evaluation. Layer 2 needs it (§5.2).

Conventions. The datum is always passed explicitly, never found by instance search, because the four types do not determine it. Besides each other and Mathlib, the eleven files import only general files, eight of them in PhyslibAlpha (1 428 lines): the extensions of the space-time algebra, multiset antidiagonals, products of representations, scalar matrices and the unit Lie algebra.

Open items. Three TODOs in the file of the structure remain, none needed for the target theorem: a constructor from a Lie group and its Lie algebra; the proof that taking Taylor series is a group isomorphism for $U(1)$ and $SU(n)$; and a constructor of `Hom` from a homomorphism of Lie groups, which embedding one gauge group in another needs.

5 Layer 2: fields

5.1 Matter fields

The structure (EXISTS). `MatterField jets` has a complex value space V, free and finite; a representation of $SL(2, \mathbb{C})$ on V; the jet action `repJet` of G_J on `SpaceTimeAlgebra` $\otimes V$, the jets of V-valued fields; the infinitesimal action `repAlgebra` of $\mathfrak{g}$ on V; a mass weight, twice the mass dimension; and a statistic, bosonic or fermionic. It has two laws. The jet action is *fibrewise*: it commutes with multiplication by scalar jets. And it is *generated* by the infinitesimal action (`IsInfinitesimalActionOf`): the Taylor coefficients of `repJet U` satisfy the two Leibniz identities determined by the Maurer–Cartan form and the adjoint coefficients of U. The second law is what makes the covariant derivative covariant.

5.2 Two more laws

The two conditions (EXISTS). Two properties hold for every physical matter field and are not implied by the structure. `PureJetsActTrivially`: a jet with trivial value acts as the identity on values at the base point; equivalently, at the base point every jet acts through its value. This is the law `evalLie_adjointJ` of the gauge datum, read on a matter field. `GaugeLorentzCompatible`: the gauge-algebra action commutes with the Lorentz action. Today they are separate definitions, assumed where covariant objects are used: 40 hypothesis binders in seven files of layer 5. The Lorentz action on the covariant algebra takes the second as an argument, and a realization of the covariant algebra carries it in its type.

The gap (EXISTS). The table layer proves `GaugeLorentzCompatible` for every row, because the two actions sit on different tensor factors. It cannot prove `PureJetsActTrivially`: `MatrixRep.matterField_pureJetsActTrivially` needs the matrix of a jet with trivial value to have identity constant term, and neither `MatrixRep` nor the factors say so. So the field data of a card has no proof of the first condition, and the hand-built Standard Model proves it species by species.

Closing it (CHECKED). State the missing property as a law of a matrix representation and, as in §4, of a factor. Then:

- `MatrixRep.trivial`, `kron`, `conj`, the charge twist of a $U(1)$ factor and the fundamental of an $SU(n)$ factor all preserve it, so the representation named by any charge tuple has it;
- the canonical factors of `ofFactors` have it;
- hence every row of every table satisfies `PureJetsActTrivially`, and the Standard Model card satisfies both conditions with no proof of its own;
- restriction along a morphism of gauge data and the direct sums `prod` and `pi` preserve both conditions.

Every way the library has of building a matter field therefore gives one that satisfies both.

Proposal (PROPOSED). Add the evaluation law to `U1Factor`, `SUFactor` and `MatrixRep`, and make the two conditions fields of `MatterField`. By rule 2 they are laws: every instance has them, and theorems should not carry them. The 40 binders go; the Lorentz action on the covariant algebra becomes unconditional; the reduction theorem (§8) needs only `Free`. The cost is two proof obligations in each construction, one line each in the prototypes, and a hypothesis on `MatterField.charged`, which nothing uses. This is the proposed answer to the question left open on 10 November. If the thread prefers hypotheses, the evaluation law is still worth adding, since without it a card cannot prove the first condition at all.

5.3 Symbols

The jet toolkit (EXISTS). V-valued jets (`Jet`), fibrewise actions on them (`JetRep`) and the Taylor coefficients `repCoeff` of a fibrewise action (`RepCoeff`, PR #2011) do not mention a gauge group. A fibrewise action is determined by its values on constant jets, a matrix of power series.

The symbol space (EXISTS). The symbols $\partial_s \psi_\alpha$ and $\partial_s \bar{\psi}_\alpha$ of a matter field M span its *jet component space*, a derivative label tensored with a covector on V, and the same for the conjugate of V. It carries the Lorentz action, the jet action, the derivative shift and the mass-weight scaling. It is contravariant in V (`JetComponentSpace.comap`), and `comap` is natural for the Lorentz action, the jet action and the scaling: three lemmas, which are all that layer 4 needs from a morphism (§7.3). A family of symbols in an algebra *transforms in* a representation (`TransformsIn`) when it transforms by the Leibniz convolution of the coefficients, with no inhomogeneous term.

5.4 Constructions and morphisms

Table rows (EXISTS). A matrix representation (`MatrixRep`) gives the action on an internal index by matrices of jets, and the infinitesimal-action law is proved once for all of them. A row of a table, a Lorentz label (`L`, `R` or `scalar`) and one charge per factor, such as `(.L, .singlet, .fund, -3)`, names a matrix representation and compiles to a matter field (`MatterFieldData.toMatterField`).

Sums (EXISTS). `prod` and `pi` are the direct sums of two matter fields and of a finite family, of a common mass weight and statistic.

Restriction (CHECKED). For a morphism of gauge data `φ : Hom jets jets'`, a matter field over `jets'` is a matter field over `jets`: the same V and Lorentz action, the jet action composed with `φ.groupJ`, the algebra action composed with `φ.lie`. The infinitesimal-action law transports, because a morphism commutes with every map the law mentions. This restricts a representation of a larger group to a smaller one, and it is the matter counterpart of `Factor.comap`.

Morphisms (AGREED 6 November). `MatterField.Hom M N` is a $\mathbb{C}$-linear map $V_M \to V_N$ commuting with the Lorentz, jet and gauge-algebra actions, between fields of equal mass weight and statistic. Nathaneal has implemented it with the notation $M \longrightarrow N$ and no category instance; the PR is not open yet. An isomorphism is an invertible morphism, an embedding an injective one, a projection a surjective one, and the structure maps of `prod` and `pi` are morphisms (TO DO).

5.5 The connection

`GaugeBoson` (EXISTS) is the value space CoVector $\otimes_{\mathbb{R}}$ $\mathfrak{g}$ of the gauge field, real, with its Lorentz action and the adjoint action of G_0. No basis of $\mathfrak{g}$ is chosen. A jet acts by $A_\mu \mapsto \mathrm{Ad}_U A_\mu + \omega_\mu(U)$: on derivative values, a linear part, the adjoint transport, plus a translation, the Maurer–Cartan coefficient. PR #2011 collects both with their composition laws in `GaugeBoson/GaugeCoeff`: the linear part composes by a Leibniz sum and the translation is a cocycle. The connection needs no morphism type. Along a morphism of gauge data its values map forward by $\mathrm{id} \otimes \varphi_{\mathfrak{g}}$ and its symbols pull back (PROPOSED, not checked).

6 Layer 3: field data

The datum (EXISTS). `GaugeFieldData jets` is a finite family of fermionic species and a finite family of bosonic species, each species a `MatterField jets` of the matching statistic. The connection is not a species: it is fixed by the gauge datum. A table (`Fields → ℕ × MatterFieldData Γ`) compiles to it with one species per field and generation.

What it derives (EXISTS). The value spaces of all fermions and of all bosons (`FermionModule`, `BosonModule`). The two *generator spaces* $E_f(T)$ and $E_b(T)$: each is the direct sum over the species of their symbol spaces, and carries the Lorentz action, the jet action, the scaling and the derivative shift, assembled species by species. The direct sum, not one symbol space on the merged value space, is what lets species carry different mass weights. With a common weight the two agree, and the merged content is one matter field (`fermionMatterField`, `bosonMatterField`).

Changes (PROPOSED, to settle when the layer starts). The rename to `FieldData`, and of the table type to `FieldTable`. The species types become parameters of the structure instead of fields: in Lean 4.35 the stored types do not reduce in concrete models, generic species lemmas fail to rewrite, and a test copy with parameters passed the same tests with no attributes (9 November). The two conditions on a datum disappear if they become laws of `MatterField`.

Morphisms. They are described with the algebra, in §7.3: species-wise morphisms and isomorphisms, built from `MatterField.Hom`, and in general maps of generator spaces.

7 Layer 4: the algebra

7.1 The algebra and its universal property

The algebra (EXISTS). Writing E_A for the real space of connection symbols,

$$J(T) = (\Lambda_{\mathbb{C}} E_f(T) \otimes_{\mathbb{C}} \mathrm{Sym}_{\mathbb{C}} E_b(T)) \otimes_{\mathbb{C}} (\mathbb{C} \otimes_{\mathbb{R}} \mathrm{Sym}_{\mathbb{R}} E_A).$$

All fermionic symbols of all species sit in one exterior algebra, so symbols of different species anticommute. The last factor is the complexified algebra of the connection alone (`LocalGaugeFieldAlgebra`), which depends only on the gauge datum.

The universal property (EXISTS). A compatible *assignment* of the generators in an associative complex algebra B, not assumed commutative, extends to exactly one algebra map $J(T) \to B$ (`Assignment.lift`, `liftEquiv`). Every action on $J(T)$ is defined as the lift of an assignment, and its laws follow from uniqueness, without unfolding the tensor product.

Realizations (EXISTS). A *realization* of T in B is an algebra map $J(T) \to B$ equivariant for the jet gauge group and the Lorentz group, both acting on B by algebra maps. Realizations are the same as assignments satisfying six generator-level laws, the law of the connection being affine (`realizationEquivAssignment`).

7.2 What the algebra carries

Actions and invariants (EXISTS). The jet gauge action and the Lorentz action; the mass-weight scaling; the weight pieces and their filtration; the submodule `invariantsLE w` of gauge and Lorentz invariants of mass weight at most w, which is what a classification describes; and `invariantsLE_map`, which carries it along an isomorphism of algebras respecting the actions and the scaling.

One grading (PROPOSED). The mass-weight grading has two encodings. $J(T)$ has the scaling maps `massWeightScale c` for real c, and the weight-n piece is their common eigenspace. The connection's algebra and the Standard Model tree have `massWeightPoly`, an algebra map $B \to B[X]$ sending a generator of weight n to X^n times itself; the classification uses it, because the weight-k part of an element is then a coefficient. Proposed: define `massWeightPoly` on $J(T)$ by the universal property and obtain the scaling by evaluating X at c, so that the weight pieces have one definition.

The total derivative (TO DO). $J(T)$ has the derivative shift on its generator spaces but not the total derivative as a derivation. The Standard Model tree has it, on the same algebra, in `JetAlgebra/JetDeriv`; that file generalises. The target theorem does not use it; invariants modulo total derivatives will.

Sectors (EXISTS). A sector is a set of categories among fermion, gauge and scalar, and its algebra is generated by the symbols of the selected categories. The classification of the Standard Model uses a different notion, the span of the words that use exactly a given set of categories. The two are reconciled when the classification is generalised.

7.3 Functoriality

$J(T)$ depends on T only through its two generator spaces with the jet action, the Lorentz action and the scaling on them. The following is CHECKED (`prototypes/Morphisms.lean`).

- **A map of generator spaces is a realization.** Linear maps $E_f(T) \to E_f(T')$ and $E_b(T) \to E_b(T')$ intertwining the three induce an algebra map $J(T) \to J(T')$ that is a realization of T in $J(T')$ and respects the scaling. Identities go to identities and composites to composites. The proof is one assignment and the existing statement that an equivariant assignment lifts to a realization.
- **An equivalence transports invariants.** If the two maps are linear equivalences, $J(T) \cong J(T')$ and the invariants of mass weight at most w of T are carried onto those of T'.
- **Species-wise isomorphisms are equivalences.** A bijection of species with isomorphisms of their matter fields gives an equivalence of generator spaces. The direct sums are reindexed with `DirectSum.lequivCongrLeft`, so no transport along equalities of species appears. So two descriptions of a theory that differ by renaming species and by isomorphisms of their matter fields have the same invariants:

```
theorem SpeciesEquiv.invariantsLE_map (S : SpeciesEquiv T T') (w : ℕ) :
    (T.invariantsLE w).map S.toGenEquiv.toAlgEquiv = T'.invariantsLE w
```

- **Species-wise morphisms go backwards.** A species-wise morphism $T \to T'$ gives a map of generator spaces from T' to T, by `JetComponentSpace.comap` and its three naturality lemmas, hence a realization of T' in $J(T)$: an expression of T' is rewritten in the fields of T.

A realization of T' in $J(T)$ is the most general way to relate two theories over one gauge datum: T' written in the fields of T. Maps of generator spaces are the linear ones, and species-wise morphisms the simplest of those. One thing is TO DO: a species-wise morphism that also commutes with the gauge-algebra actions, as `MatterField.Hom` does, carries covariant towers to covariant towers, by the naturality of the covariant derivative.

8 Layer 5: the covariant algebra

The calculus (EXISTS). In any algebra carrying the connection, the field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]$ and the covariant derivative $\nabla_\rho F = \partial_\rho F + A_\rho \cdot F$ of a family that transforms in a representation, through the infinitesimal action. The covariant derivative preserves `TransformsIn`; this is where the second law of a matter field is used. The Lorentz law of the covariant towers is proved by one induction.

The covariant algebra (EXISTS). In $J(T)$, the covariant towers $\nabla_{l_1} \cdots \nabla_{l_n} \psi$ of every species and of their conjugates, and the field strength with its covariant derivatives. `LocalCovFieldAlgebra` is the subalgebra they generate. A jet acts on a tower through its zeroth Taylor coefficient on the value index alone. With the first condition of §5.2 that is the action of its value, so on this subalgebra the jet action factors through G_0 and gauge invariance is ordinary invariant theory. The Lorentz group preserves the subalgebra under the second condition.

The reduction theorem (CHECKED, `prototypes/Covariance.lean`). For free gauge data and species satisfying the first condition, an element of $J(T)$ is fixed by every jet of gauge transformations exactly when it lies in the covariant algebra and is fixed by the constant jets:

```
theorem forall_repJet_eq_iff [jets.Free] (hP : T.PureJetsActTrivially)
    (x : T.LocalFieldAlgebra) :
    (∀ U : GJ, T.repJet U x = x)
      ↔ x ∈ T.LocalCovFieldAlgebra ∧ ∀ g : G₀, T.repJet (jets.ofConstant g) x = x
```

The hard part was already general: the reduction for the gauge field in the presence of further generators fixed by the pure jets (`invariant_mem_adjoin_fieldStrength`), which uses `Free`, and the span lemma that exchanges the derivative symbols of a matter family for its covariant tower. What was missing is that the gauge-field symbols and the matter towers generate $J(T)$, and the assembly, about 230 lines. The Standard Model tree proves the same statement for its twelve named families. With the evaluation law the theorem applies to every card; the prototype states it for the Standard Model card (`StandardModel.Model.forall_repJet_eq_iff`), with no proof specific to the model.

9 The Standard Model

9.1 What exists

The classification is proved (EXISTS). After the reduction theorem, an invariant is an element of the covariant algebra fixed by G_0 and by the Lorentz group. The covariant algebra is cut into sectors by which classes of field a word uses, and each sector is classified at each mass weight up to 8 with the invariant theory of $SU(3)$, $SU(2)$, $U(1)$ and the Lorentz group. The pieces are joined with

`ReducesInvariantsTo`: a reduction of V to W says that for every stable submodule S, an invariant of $V \sqcup S$ lies in $W \sqcup S$. Reductions compose and combine over joins, so the sectors are taken one at a time and their independence is never needed. The result, `mem_massWeightSubmoduleLE_eight_and_invariant_iff_lagrangian`, is stated for every algebra that realizes the Standard Model, and for its own algebra with no side condition: an element of mass weight at most 8 is invariant exactly when it is a combination of the constant, $H^\dagger H$, the gauge kinetic and theta terms, the Higgs kinetic, quartic and box terms, the fermion kinetic terms and the Yukawa couplings.

9.2 Two towers

The general layers 4 and 5 were written after the Standard Model tree and generalise it, but only the algebra has been switched over: `JetAlgebra` is `StandardModel.fieldData.LocalFieldAlgebra`, for a hand-built datum with the fifteen fermion multiplets and the Higgs. Everything else exists twice.

	Standard Model tree	general layers
algebra	`JetAlgebra`	`T.LocalFieldAlgebra` (the same)
actions	`repJetGaugeGroupI`, `repLorentzGroup`, tensor products of the actions on the factors	`T.repJet`, `T.repLorentzGroup`, lifts by the universal property
grading	`massWeightPoly`	`massWeightScale`
realization	`AlgebraRealization`	`T.Realization`
covariant algebra	`covAlgebra`, `CovJetAlgebra`, `CovAlgebraRealization`	`T.LocalCovFieldAlgebra` and its realizations
reduction	`forall_repJet_eq_iff`, twelve families	§8, prototype
classification	sector structures, the spans, the theorem	none

Both columns rest on the same calculus in an arbitrary algebra: the symbol spaces, the covariant derivative and the reduction for the gauge field. Beyond that, the Standard Model tree does not use the general actions, grading, realizations or covariant algebra. The target theorem is therefore a statement in the left column, about the hand-built datum.

9.3 The route to the card

1. **Compare the two columns on the shared algebra** (TO DO). The two jet actions agree, and the two Lorentz actions, because each pair consists of algebra maps that agree on the generators. The two gradings give the same weight pieces: the scaling is the polynomial evaluated at c, and a polynomial is determined by its values. The theorem then reads: `StandardModel.fieldData.invariantsLE 8` is the span of the list.
2. **Show the two descriptions equivalent** (mechanism CHECKED, the isomorphisms TO DO). The hand-built datum and the card's `StandardModel.Model.fieldData` have the same gauge datum, by `rfl`. A species-wise isomorphism between them is a bijection between the fifteen hand-built species and the card's species, each a field together with a generation, and an isomorphism of matter fields for each of the five fermion types and for the Higgs. For the lepton doublet the two matter fields are equal by `rfl`; for the quark doublet the matrices are shown to agree. By §7.3 the invariants of the card are then the image of the list. This is the first use of morphisms, and it is what Joseph asked for on 4 November.
3. **Read the list on the card** (TO DO). The theorem holds in every realization of the Standard Model, and the equivalence makes the card's algebra one. The spans of the list are then spans of the card's own covariant towers, since the equivalence carries towers to towers (§7.3).
4. **Afterwards: one tower.** Replace the structures of the left column by those of the right, starting with the reduction theorem, and state the sector classifications for the general covariant algebra.

Steps 1 and 2 give the target theorem on the card. The table shows what each layer contributes to it.

layer	what the theorem on the card uses	status
1	`ofFactors [.SU 3, .SU 2, .U1]` is free	EXISTS
2	the six rows are matter fields satisfying both conditions	CHECKED
3	the table compiles to field data, 15 fermionic species and one bosonic	EXISTS
4	J of the card with its actions, grading and `invariantsLE`; equivalent descriptions have the same invariants	EXISTS; CHECKED
5	on the card, gauge invariance is covariance plus global invariance	CHECKED
classification	the list, proved on the hand-built datum	EXISTS; steps 1, 2

10 Relating theories

what is wanted	mechanism	status
Two descriptions of one theory are the same	a species-wise isomorphism; it induces $J(T) \cong J(T')$ and carries invariants to invariants	CHECKED
One multiplet inside another; generations as copies of one field	`MatterField.Hom`	AGREED; implemented, PR not open
	the structure maps of `prod` and `pi` are morphisms	TO DO
	a species-wise morphism $T \to T'$ realizes T' in $J(T)$	CHECKED
Flavour rotation, change of Higgs basis	a map of generator spaces that mixes species of one type; it is a realization like any other	CHECKED; its constructor from a matrix TO DO
A smaller gauge group; one factor of a product	`LocalGaugeData.Hom`, `Hom.fst`, `Hom.snd`, `Factor.comap`	EXISTS
	restriction of matter fields, `MatterField.comap`	CHECKED
	the connection: values map forward, symbols pull back	PROPOSED
One theory written in the fields of another	a realization of T' in $J(T)$	notion EXISTS

Two remarks. Over one gauge datum the automorphisms of the generator spaces of the Standard Model fermions include the flavour group, which species-wise isomorphisms do not see; that is why the general notion is a map of generator spaces and the species-wise one only its simplest case. And the branching of a gauge sector, such as the X and Y bosons of $SU(5)$ seen from the Standard Model, is a statement about algebras, not about field data: the larger connection restricted to the smaller group is the smaller connection together with a vector field that transforms linearly, which needs a vector label and real fields.

11 Decisions and plan

Agreed. Matter and the connection are two separate notions (2–3 November). The species type of PR #2011 is dropped, and layers 3 to 5 are named `FieldData`, `LocalFieldAlgebra/` and `LocalCovFieldAlgebra/` (3 November). Morphisms of matter fields are `MatterField.Hom`, with the quiver notation and no category instance (6 November). The work goes layer by layer, layer 1 by Jinzheng and layer 2 by Nathaneal (10 November).

For the thread to decide.

question	proposal	evidence
The rest of PR #2011	Keep `RepCoeff`, `InfinitesimalAction/Basic` and `GaugeCoeff`; narrow three imports.	§3, CHECKED
The layers	The assignment of §3: the connection's algebra and the covariant derivative are parts of layers 4 and 5.	import check
The two conditions	An evaluation law on `U1Factor`, `SUFactor` and `MatrixRep`; then the two conditions become fields of `MatterField`.	§5.2, CHECKED
Restriction	`MatterField.comap` along `LocalGaugeData.Hom`.	CHECKED
Field data	Species types as parameters; the renames. To settle when layer 3 starts.	tests of 9 Nov
Morphisms of theories	Maps of generator spaces, the species-wise ones first; in general a realization.	§7.3, CHECKED
The grading	`massWeightPoly` on $J(T)$, the scaling derived from it.	§7.2
The target theorem	On the card, by steps 1 and 2 of §9.	§9

Layer 1 (Jinzheng, upstream to `Physlib/`).

1. The consequences of the laws and `Faithful`, with `MultisetAntidiagonal`; then `AdjointCoeff` and `MaurerCartan`; then `Truncation`, which brings `Free`. These four files (1 214 lines) need no other prerequisite.
2. With Nathaneal, who wrote the space-time algebra in `Physlib/`: the four files extending it. Then `Factor`, `MatrixJets`, `Prod`, `SU/Algebra`, `SU/Basic`, `U1`, `OfFactors`, and the three small general files.
3. If the thread accepts it, the evaluation law on the factors, as a small PR in PhyslibAlpha now, so that layer 2 does not wait for the upstreaming.
4. The $SU(n)$ invariant theory stays in PhyslibAlpha.

Layer 2 (Nathaneal, in PhyslibAlpha).

1. Rework PR #2011 as above; narrow the three imports; move the jet toolkit below the layer; split the model table from the rows.
2. `MatterField.Hom` with its notation.
3. Decide the two conditions. If they become laws: the evaluation law on `MatrixRep`, then the two fields. The prototypes contain every proof needed.
4. Restriction, from the prototype.

Done when no file of the layer imports anything above it.

Layers 3 to 5 and the theorem (after layers 1 and 2).

1. Layer 3: the renames; species types as parameters.
2. Layer 4: functoriality, from the prototype; one grading; the total derivative.
3. Layer 5: the reduction theorem, from the prototype; the hypotheses removed.
4. The theorem on the card: steps 1 and 2 of §9.

The reduction theorem and functoriality do not depend on the decisions above and can be opened as PRs in PhyslibAlpha at any time.

12 After the deadline

- **Reality.** Every matter field is complex and the conjugate symbols are already in its symbol space. A real scalar or a Majorana fermion cannot be expressed yet. Proposed: a conjugate-linear involution of V commuting with the actions, as separate data on a matter field, and a star on $J(T)$; a real Lagrangian is then a star-fixed invariant.
- **Total derivatives and equations of motion.** With the total derivative on $J(T)$, invariants modulo its image. Rule 4 is what makes this cheap: the classification is stated in any realization, and a quotient is one.
- **One tower** for the Standard Model (step 4 of §9), and the sector classifications stated for a general field datum.
- **The card.** To write a Lagrangian from it, a card needs a way to write a term from its symbols and check its invariance, a gauge field and a field strength named for each factor, a user-facing conjugation, real fields and a vector label.
- **Extensions.** The sum of two field data, with the invariants of an extension reduced to its mixed part (a TODO in `LocalField/Sector`); symmetry breaking; higher mass dimension.

A The prototypes

Three files in `prototypes/` of this repository, compiled against the branch of PR #2011 (commit `800e1cf6`, Lean v4.35.0) by running `lake env lean` on each file from a Physlib checkout. They are checks of the design, not code for review; each can become a PR.

file	lines	what it proves
`MatterFields.lean`	154	`MatterField.comap`: restriction along `LocalGaugeData.Hom`, with the infinitesimal-action law transported. The two conditions are preserved by `comap`, `prod` and `pi`.
`Covariance.lean`	499	The evaluation law (`EvalLaw`) on factors and matrix representations; it is preserved by every construction of the table layer and holds for the canonical factors, so every table row satisfies `PureJetsActTrivially`. The gauge-field symbols and the matter towers generate $J(T)$. The reduction theorem `mem_localCovFieldAlgebra_of_gaugeInvariant` and `forall_repJet_eq_iff`. The same for the Standard Model card.
`Morphisms.lean`	562	`GenHom`: a map of generator spaces induces a realization, functorially. `GenEquiv`: an equivalence induces $J(T) \cong J(T')$ and `invariantsLE_map`. `SpeciesEquiv`: a species-wise isomorphism is an equivalence of generator spaces. `SpeciesHom`: a species-wise morphism $T \to T'$ realizes T' in $J(T)$.

A practical remark: the generator spaces are definitions, not abbreviations, so in Lean 4.35 lemmas about `DirectSum` do not apply to them directly. Their own interface, `assembleFermion`, `inclFermion` and `fermionGenerators_hom_ext` and the bosonic versions, is enough.

B History

PR #1415 (September 2026) made layer 1 independent of matter and introduced `MatrixJets`; `MatrixJets.free` (Joseph, 25 September) reduced freeness to conditions on the carriers. The tree moved to PhyslibAlpha in PR #1801, and `MatterField` gained its statistic in PR #1989 (30 October). PR #2011 introduced a species type, dropped on 3 November. Maps between theories were requested on 4 November, and `MatterField.Hom` was agreed on 6 November. On 10 November the thread returned to the layer order; this version reviews the code against the design and checks its proposals.